

Practical - 10: Data Transformation and Standardization for Data Analysis

Introduction

Data transformation and standardization are important steps in data analytics. Raw data often comes in different formats, which are not suitable for direct analysis. This practical demonstrates how to convert and standardize data into a structured format using Python.

Objective

The objective of this practical is to understand data transformation techniques, convert raw data into structured format, and prepare datasets for analytics and machine learning.

Tools and Technologies

This practical uses Python programming in Jupyter Notebook. Libraries used include Pandas for data handling and Scikit-learn for scaling numerical values.

Dataset Description

The dataset used is `employee_data.csv` which contains columns such as ID, Name, Age, Salary, and Join_Date. The dataset requires transformation to make it suitable for analysis.

Step 1: Import Libraries

Required libraries such as `pandas` and `StandardScaler` from `sklearn` are imported to perform data manipulation and scaling.

Step 2: Load Dataset

The dataset is loaded using `pandas read_csv()` function. The `head()` function is used to display the first few rows.

Step 3: Convert Date Format

The `Join_Date` column is converted into datetime format using `pd.to_datetime()`. This ensures consistency in date representation.

Step 4: Convert Categorical Data

Text data such as Name is converted into string type using `astype()`. This helps in proper data handling.

Step 5: Standardize Numerical Data

Numerical values such as Salary are scaled using `StandardScaler`. This process normalizes the data, making it suitable for machine learning algorithms.

Step 6: Verify Transformed Data

The transformed dataset is checked using `head()` to verify changes.

Step 7: Save Processed Dataset

The final dataset is saved into a new CSV file using `to_csv()` for further use.

Advantages of Data Transformation

Data transformation improves data quality, ensures consistency, enhances model performance, and enables better analysis.

Conclusion

This practical demonstrates how raw data can be transformed into a standardized format. It is an essential step in data preprocessing and analytics workflows.

Outcome

After completing this practical, students can perform data transformation, standardization, scaling, and prepare datasets for machine learning and analytics.